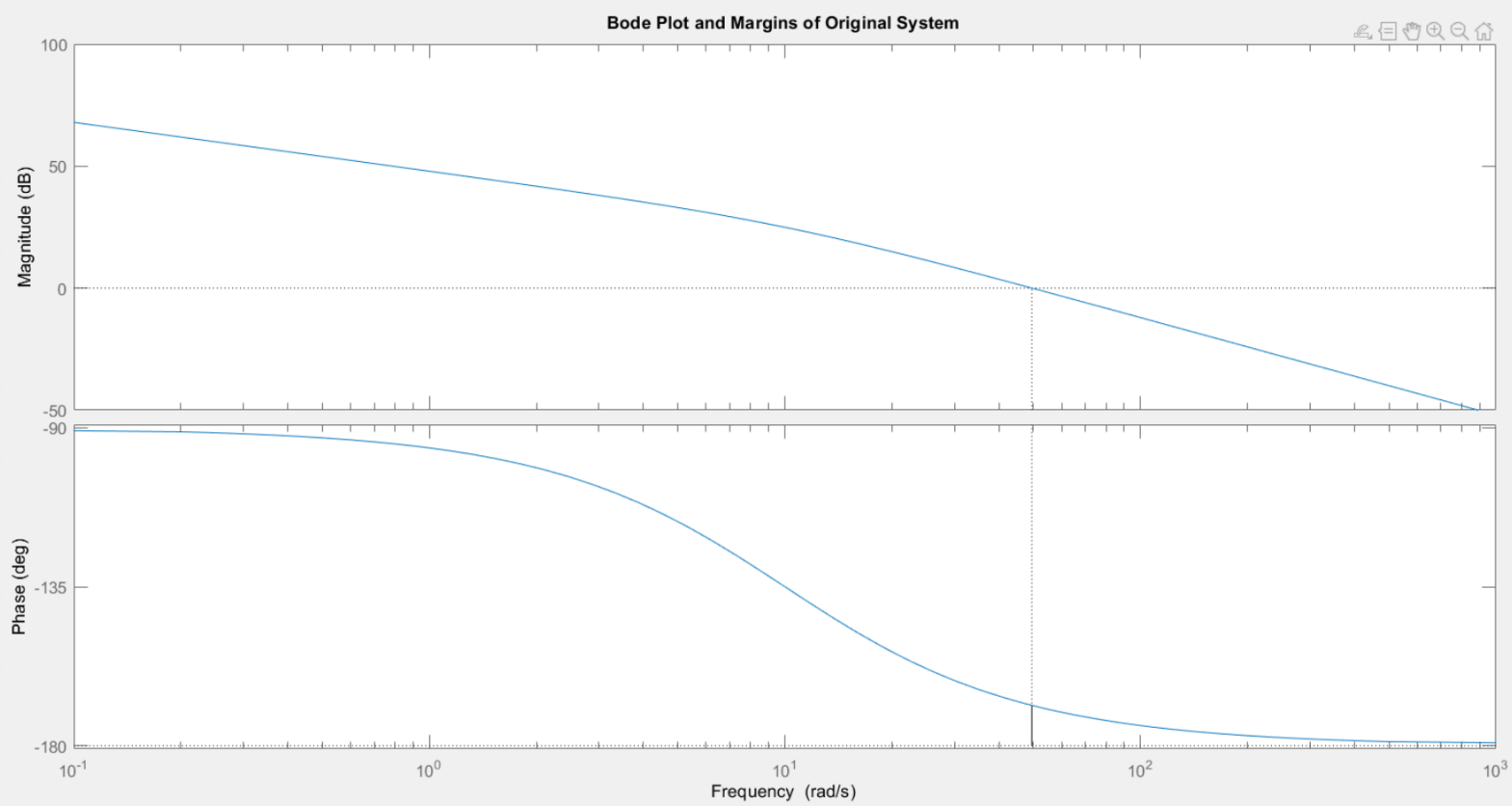


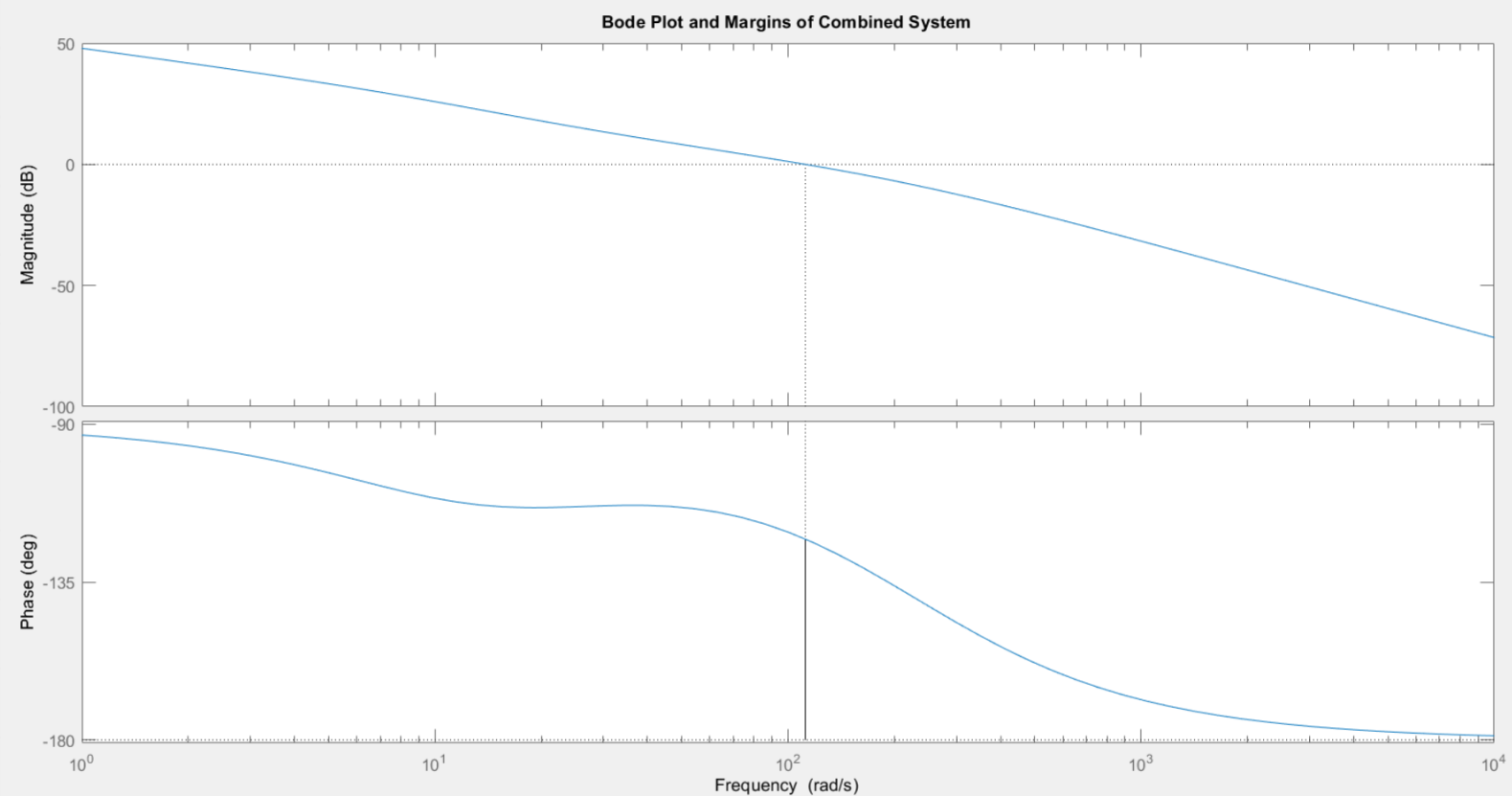
Problem 1

$$G(s) = \frac{250}{s(0.1s+1)}$$

$$G(s)G_c(s) = \frac{250}{s(0.1s+1)} \frac{0.015s+1}{0.0017s+1}$$



Cutting frequency $\omega_{co} = 49.5012$ rad/s
 Phase margin $\gamma_0 = 11.42^\circ$



Cutting frequency $\omega_{ec} = 111.7308 \text{ rad/s}$
Phase margin $\gamma_c = 57.23^\circ$

i) The phase-lead compensator improves the phase margin γ by offsetting the negative phase introduced by original $G(s)$, stabilizing the system, reducing overshoot.

ii) It also increases the bandwidth of system by pushing cutting frequency to a higher value, thus improving the speed of system's response. (short t_s).

Problem 2

$$G_0(s) = \frac{100}{s(10s+1)}$$

$$G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s}$$

$$G(s) = G_0(s) \cdot G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s} \cdot \frac{100}{s(10s+1)}$$

$$= \frac{100 K_D s^2 + 100 K_P s + 100 K_I}{10s^3 + s^2}$$

$$\text{CLTF } B(s) = \frac{G(s)}{1 + G(s)}$$

$$= \frac{100 K_D s^2 + 100 K_P s + 100 K_I}{10s^3 + s^2}$$

$$1 + \frac{100 K_D s^2 + 100 K_P s + 100 K_I}{10s^3 + s^2}$$

$$= \frac{100 K_D s^2 + 100 K_P s + 100 K_I}{10s^3 + s^2} \times \frac{10s^3 + s^2}{100 K_D s^2 + 100 K_P s + 100 K_I + 10s^3 + s^2}$$

$$= \frac{100 K_D s^2 + 100 K_P s + 100 K_I}{10s^3 + (100 K_D + 1)s^2 + 100 K_P s + 100 K_I}$$

$$10s^3 + (100 K_D + 1)s^2 + 100 K_P s + 100 K_I = 0$$

$$[s - (-2+j1)][s - (-2-j1)][s - (-5)]$$

$$= (s+2-j1)(s+2+j1)(s+5)$$

$$= s^3 + 9s^2 + 25s + 21$$

$$10K_D + \frac{1}{10} = 9 \quad 10K_P = 25 \quad 25 = 100K_I$$

$$K_D = 0.89 \quad K_P = 2.5 \quad K_I = 0.25$$

$$\text{PID controller TF } G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s} = \frac{0.89s^2 + 2.5s + 0.25}{s}$$

Problem 3

$$3.1 a) \quad G_o(s) = \frac{K_m}{s(7ms+1)} \quad G(s) = G_c(s) G_o(s) = \frac{(K_P + K_D s) K_m}{s(7ms+1)}$$

$$B(s) = \frac{\frac{(K_P + K_D s) K_m}{s(7ms+1)}}{1 + \frac{(K_P + K_D s) K_m}{s(7ms+1)}}$$

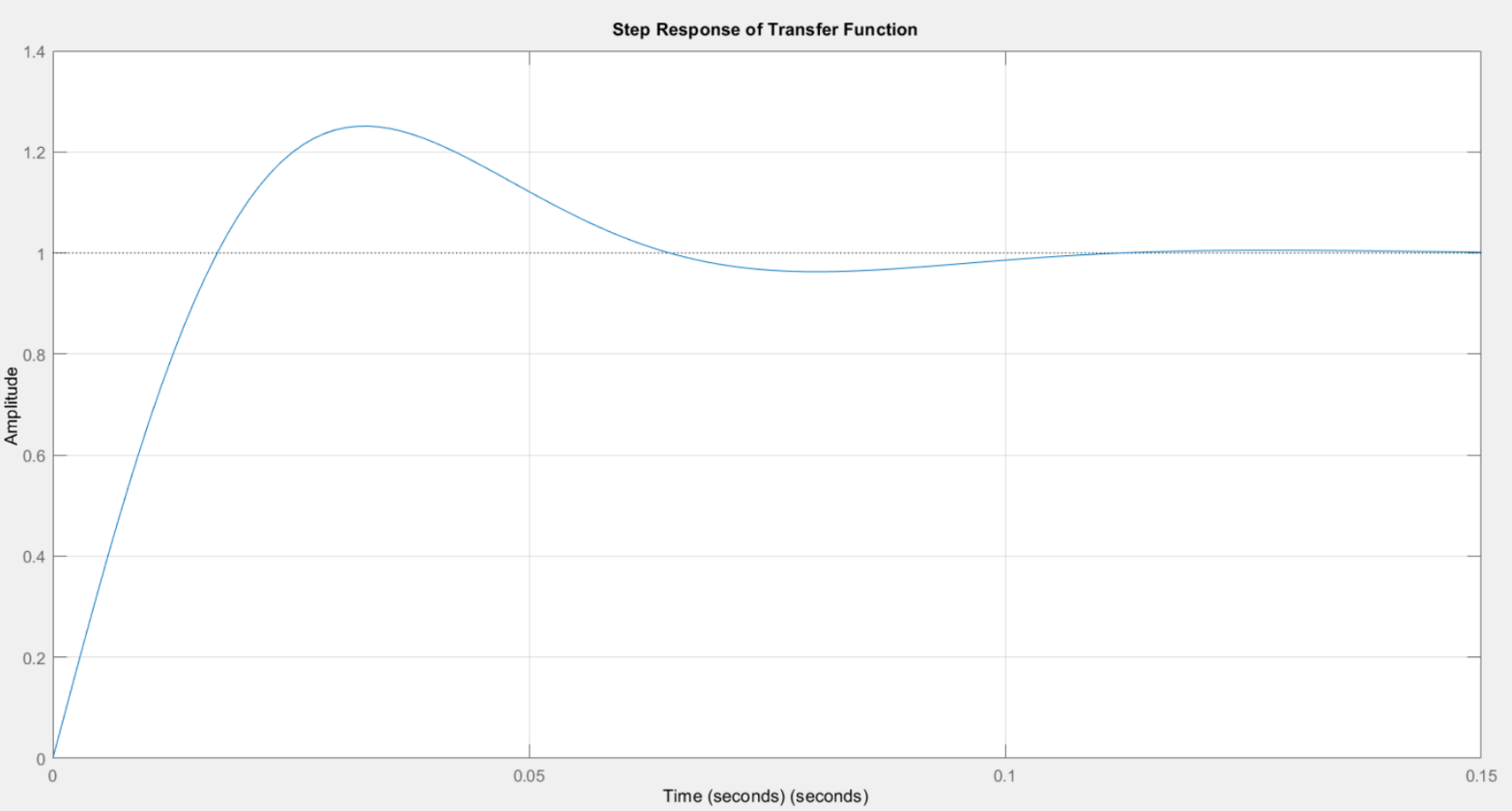
$$= \frac{(K_P + K_D s) K_m}{(K_P + K_D s) K_m + s(7ms+1)}$$

$$= \frac{(4.08 + 0.048s) 147}{(4.08 + 0.048s) 147 + s(0.15s+1)}$$

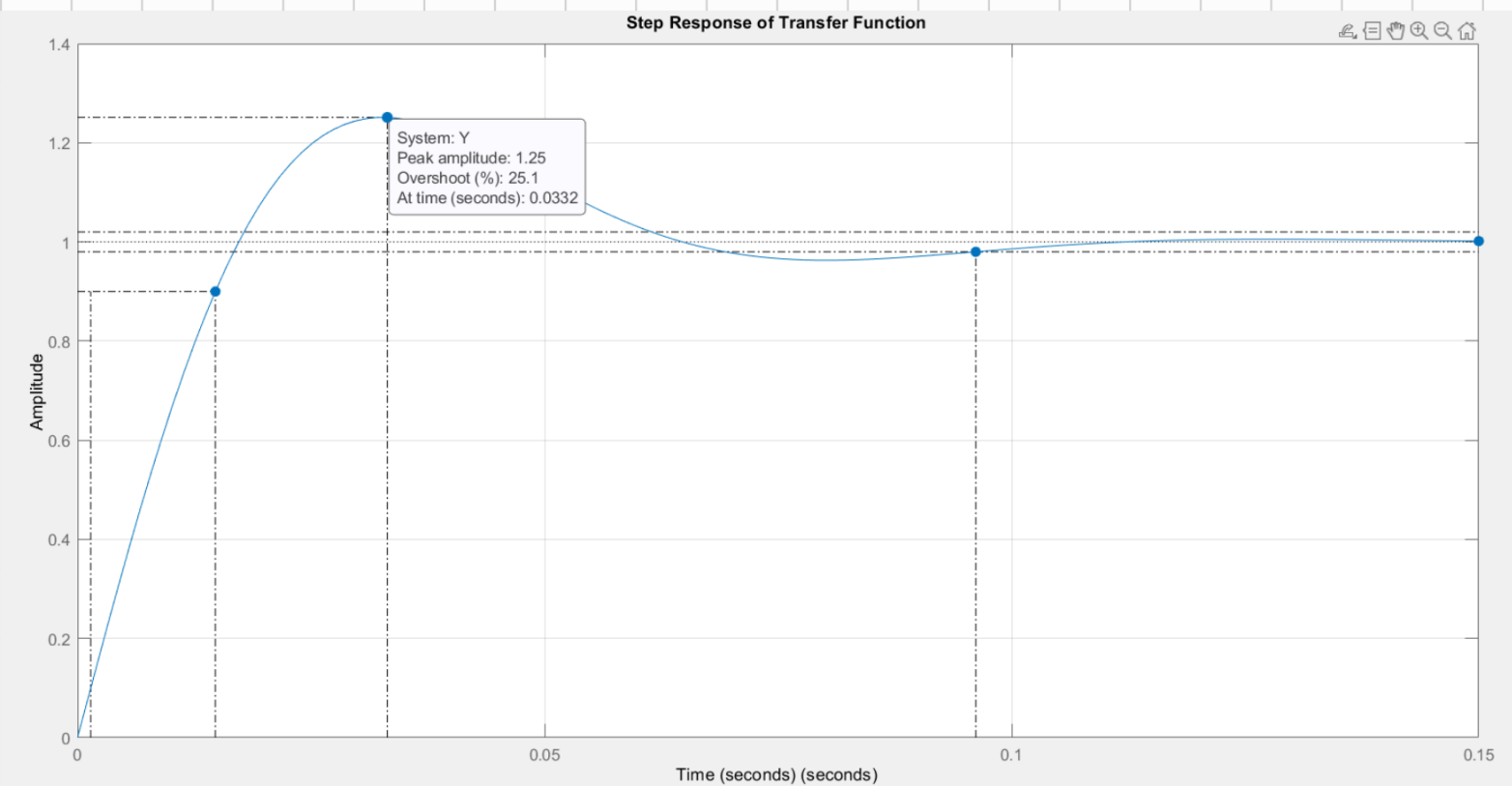
$$= \frac{599.76 + 7.056s}{599.76 + 7.056s + 0.15s^2 + s}$$

$$= \frac{7.056s + 599.76}{0.15s^2 + 8.056s + 599.76}$$

$$= \frac{7.056s + 599.76}{s^2 + 80.56s + 5997.6}$$



3-26) We obtain plot of step response of DC motor drive system with Overshoot of 25.1 %



2% settling time of 0.1 sec

$$t_s = \frac{4}{\delta} < 0.1$$

$$\delta > 40$$

$$\%OS < 15\%$$

$$\bar{\varphi} = \tan^{-1} \left[-\frac{\pi}{\ln(15\%)} \right] = 58.87^\circ$$

$$\text{poles}_{inv} = -40.3 \pm 66.1$$

$$\text{zero} = -85$$

$$G_f(s) = \frac{1}{s+85}$$

$$G(s) = G_c(s) G_o(s) = \frac{(K_p + K_D s) K_m}{s(2ms+1)}$$

$$G_f(s) \cdot G_c(s) G_o(s) = \frac{59976}{(7.056s + 599.76)(s+100)} \cdot \frac{(4.08 + 0.00485s)147}{s(0.1s+1)}$$

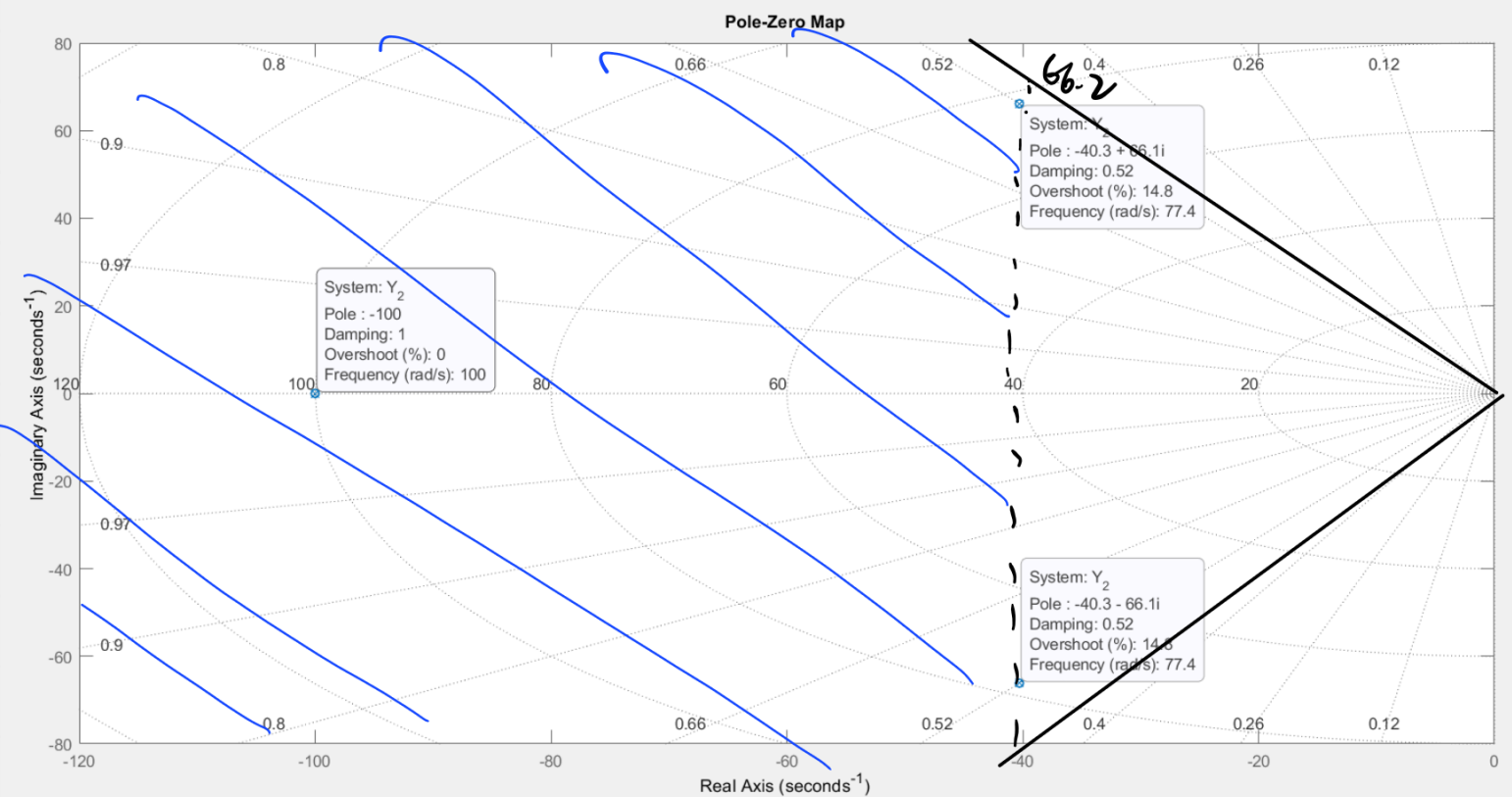
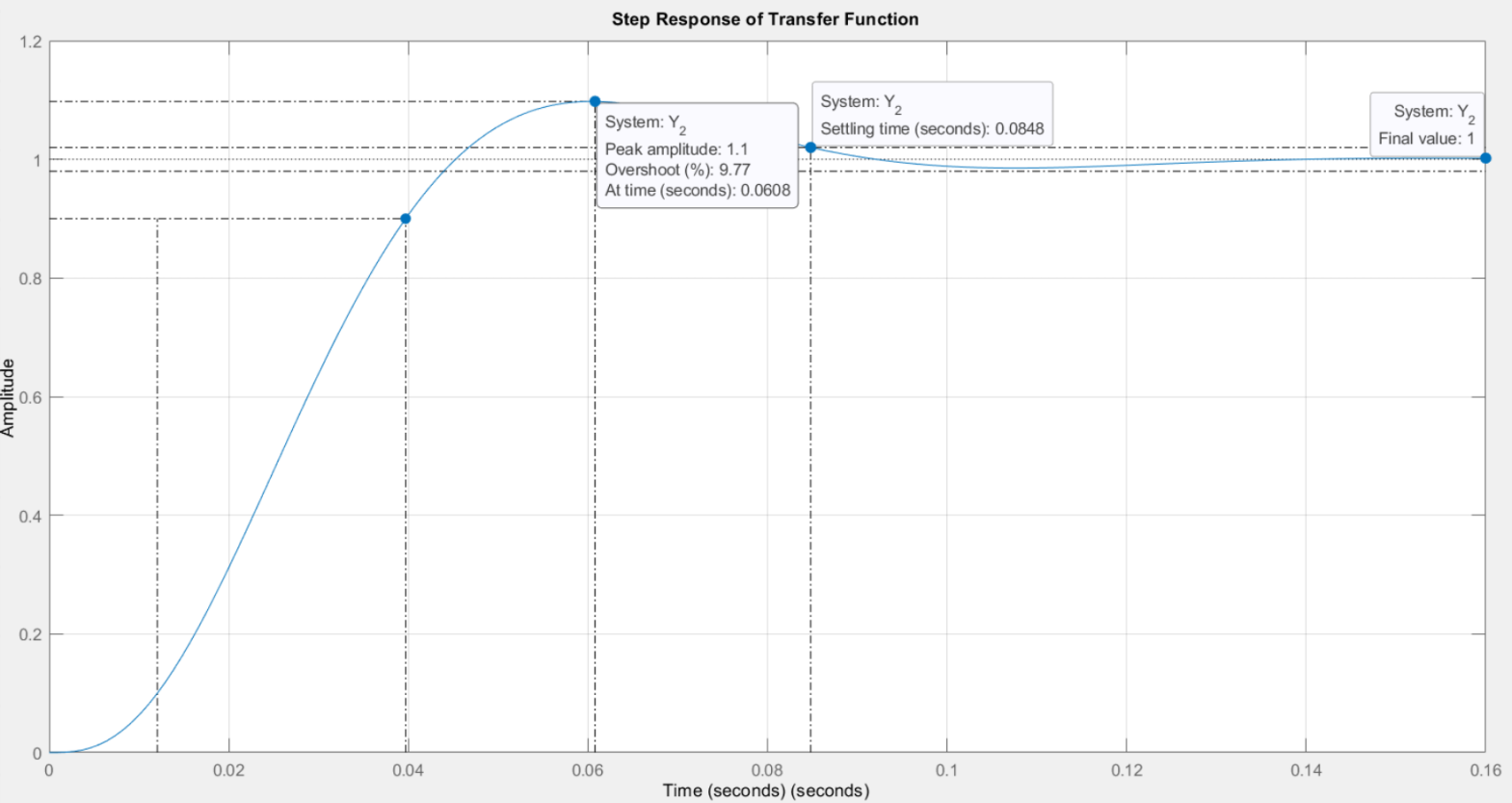
$$B_2(s) = \frac{59976}{s+100} = \frac{59976}{0.1s^3 + 18.056s^2 + 1405.36s + 59976}$$

$$1 + \frac{7.056s + 599.76}{s(0.1s+1)}$$

$$G_f(s) = \frac{59976}{7.056s + 599.76(s+100)}$$

since the overshoot is contributed by the additional zero $(7.056s + 599.76)$
 $s+100$ but 599.76 is also raised by 10^2
 so steady state value is preserved as 1.

$$\%OS = \frac{1}{\sqrt{1-\zeta^2}} = 1.77 \Rightarrow \zeta = 0.9947$$



As you can see %Overshoot is 9.77% < 15% , % settling time $t_s = 0.085 < 0.1$ s and final value is 1

All poles fulfil the requirement that their real parts < -40 and imaginary parts are within the blue area.